



PROBLEM SOLVING WITH C

UE25CS151B

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PROBLEM SOLVING WITH C

Language Specifications/Behaviors



1. Language Specification
2. Standards
3. Behaviors defined by C Standards
4. Undefined Behavior in detail

Language Specification

- A documentation that defines a programming language so that users and implementers can agree on what programs in that language mean.
- Are typically detailed and formal, and primarily used by implementers referring to them in case of ambiguity.
- Can take several forms:
 - An explicit definition of the syntax and semantics of the language.
 - A description of the behavior of a "translator" for the language
 - "Model implementation" is a program that implements all requirements from a corresponding specification

Standards

de Facto: Practices that are legally recognized, regardless of whether the practice exists in reality.

de Jure: Describes situations that exist in reality, even if not legally recognized

- Language may have one or more implementations which acts as deFacto Language may be implemented and then specified, or vice versa or together.
- Languages can exist and be popular for decades without a specification - Perl
- After 20 years of usage, specification for PHP in 2014
- ALGOL 68 : First (and possibly one of the last) major language for which a full formal definition was made before it was implemented

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Language Specifications/Behaviors



Behaviors defined by C Standards

- Locale-specific behavior - Not discussed here
- Unspecified behavior - Order of evaluation of arguments in printf function
- Implementation-defined behavior – Size of each type
- Undefined behavior in detail

Undefined Behavior in detail

- The result of executing a program whose behavior is prescribed to be unpredictable in the language specification to which the computer code adheres.
- It is the name of a list of conditions that the program must not meet.
- Examples: Memory access outside of array bounds, Signed integer overflow
- Standard imposes no requirements: May fail to compile, may crash, may generate incorrect results, may fortunately do what exactly programmer intended
- Coding Examples



THANK YOU

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